

Thumbnail Guided Tissue Compaction (TGTC) - Whole Slide Image Preprocessing for Lossless Image Compression

Christopher Smith, Leo Galway, Richard Gault

Centre for Intelligent Sustainable Computing, School of Electronics, Electrical Engineering, and Computer Science, Queen's University, Belfast

Abstract

Whole-slide images (WSI's) of histopathology slides are difficult to store and transmit because scans routinely occupy multiple gigabytes before compression. While some details of the image must never be lost due to their clinical significance, there are frequently background details that are less relevant. This paper proposes Thumbnail-Guided Tissue Compaction (TGTC) that detects tissue on a low-resolution thumbnail of the sample before a crop-and-rotate calculation is applied. This acts as a preprocessing stage to existing image compression algorithms aiming to optimally position the sample to minimise background information. TGTC isolates and extracts the important tissue details and reduces the amount of redundant background information being stored in the compressed file. This study considers 12 WSI's captured from a diverse range of pathologies. On average, TGTC reduces the storage constraints by 21.35% and up to 69.91% in some cases. A thumbnail-level retention audit evaluated the efficacy of identifying tissue with such a low-resolution image and found a mean retained tissue fraction 99.94%. Across 48 paired slide-method comparisons, TGTC reduces final compressed size in 40/48 cases, with mean improvement 10.39% and mean codec-ratio delta of 0.71. These results support TGTC as a practical WSI preprocessing stage when the goal is reduced final storage with an explicit tissue-preservation audit trail.

Keywords: whole-slide imaging, lossless compression, digital pathology, preprocessing, tissue detection.

1 Introduction

Whole-slide imaging is central to digital pathology, but its storage demands remain a practical barrier for archival use, dataset sharing, and large-scale computational pathology studies. Studies found that the average storage cost for digital pathology labs across seven sites in the European Union is €155,000 per site annually [1]. A level-0 RGB representation of a single slide can occupy several gigabytes before compression, and benchmarking many such slides across multiple codecs quickly becomes expensive in both runtime and disk use. This work studies a simple idea: rather than compressing the full slide canvas, first identify the tissue footprint from a thumbnail, estimate a tight rotated bounding region, and then compress only the resulting tissue-focused crop. The proposed Thumbnail-Guided Tissue Compaction (TGTC) pipeline is designed to answer two questions at once. First, does TGTC reduce the number of bytes that must ultimately be stored? Second, can that reduction be defended as effectively lossless with respect to tissue retention?

The answer is not obvious. A smaller image does not automatically imply a better codec ratio (or compression ratio) because the transformed image may be less compressible per byte and the distinction between these two goals is important. The downstream codec stage can be tested for strict bitwise losslessness tile by tile. The TGTC preprocessing stage cannot make the same claim in original slide coordinates because it changes orientation and canvas extent. Instead, it should be assessed through an explicit tissue-retention audit. This paper therefore separates raw size reduction, codec ratio, final compressed size, timing cost, and tissue-retention evidence rather than treating compression ratio alone as the sole objective.

The main contributions of the study are:

1. A novel pre-processing stage to extend existing lossless compression methods that reduces redundant high-fidelity compression seen in existing methods.
2. A reviewer-facing definition of tissue-preservation based on retention-audit evidence rather than pixel-wise preservation after crop-and-rotate preprocessing, as rotation inherently changes pixel coordinates.
3. A storage-centric analysis showing that TGTC often reduces final compressed size even if the transformed image exhibits a lower codec ratio than the baseline canvas.

2 Literature Review

WSI's can reach gigapixel scales meaning efficient compression is hugely important for the practical utilisation of these data. The literature shows a clear progression from general-purpose algorithms to specialised, high-performance tools for WSI compression. Welch [2] detailed the LZW (Lempel-Ziv-Welch) algorithm that became a standard for lossless data compression (notably in GIF and TIFF formats). While reliable, LZW often struggles with the high-entropy demands of modern high-resolution imaging. Collet and Kucherauw [3] standardised the Zstandard (Zstd) algorithm in 2021. Zstd offers a balanced performance profile, delivering compression ratios comparable to traditional dictionary-based methods like DEFLATE and zlib, alongside significantly higher decompression speeds suitable for real-time data streaming in digital systems. Alakuijala et al. [4] introduced JPEG-XL, designed specifically for the requirements of high-fidelity images. It offers superior compression efficiency and features - such as lossless transcoding of legacy JPEGs and responsive web delivery - that are particularly relevant for the long-term storage and remote viewing of medical slides.

Mao et al. [5] introduced the WISE framework, a classical, non-learned approach to lossless compression of gigapixel whole-slide images. WISE combines hierarchical predictive transforms, bitplane decomposition, and dictionary-based entropy coding, motivated by the observation that histopathology slides exhibit significant spatial redundancy and background sparsity. Operating on tiled representations, the framework is designed to scale to gigapixel data without requiring full-resolution processing. More broadly, WISE highlights the importance of content-aware redundancy in gigapixel imagery, motivating the exploration of adaptive, learned compression approaches considered in this study.

While conceptually grounded in established lossless compression techniques, WISE is presented at a high level and does not specify several implementation-critical details, such as residual representation, bitplane encoding, and entropy coding constraints. In the absence of an official reference implementation, exact reproducibility is therefore not guaranteed. In this study, a faithful implementation approximation of WISE is included based on the available description and its use as one of several comparative baselines. Nevertheless, the framework provides a useful reference point for assessing redundancy reduction strategies in WSI's.

3 Experimentation

3.1 Data

A selection of 12 whole slide images were extracted from The Cancer Genome Atlas (TCGA) [6] in their original .svs proprietary file format to represent 12 different potential cancerous sites of the human body; specifically adrenal gland, bladder, brain, breast, cervix, colon, heart, kidney, lung, ovary, pancreas, and skin. These sites were chosen to exemplify the wide variability of background content found in different histopathologies.

3.2 Thumbnail-Guided Tissue Compaction (TGTC)

The proposed pre-processing block, TGTC, comprises of 3 core steps:

1. foreground/background separation with Otsu Thresholding [7],
2. the isolation of biological tissue within the image, and

3. rotation & cropping to produce a minimal tissue region along the angle of inclination, where the images sides run along the x- and y-axis.

Each source WSI `.svs` file is opened with OpenSlide and read at level 0. The baseline export streams RGB tiles into memory-mapped raw unsigned 8-bit integers (`uint8`). The resulting rectangle is mapped back into full-resolution coordinates. A perspective transform is then used to stream the rotated crop, tile by tile, into a new `.dat` file where it is then saved. Each slide has a baseline memory-mapped RGB `.dat` representation stored in `uint8` and a TGTC-transformed memory-mapped RGB `.dat` representation created from the original `.svs` file. This tiled extraction avoids creating a massive intermediate rotated image in memory and makes the method feasible on large WSI canvases.

3.3 Compression Methods

Four lossless compression methods are benchmarked based on the review of the existing literature, namely:

1. **LZW**: dictionary-based lossless compression in the spirit of the classic LZW formulation [2].
2. **Zstandard**: byte-oriented entropy coding using Zstandard at compression level 3, the default setting [3].
3. **JPEG-XL (lossless)**: lossless still-image compression via JPEG-XL [4].
4. **WISE (Custom)**: a custom three-stage pipeline consisting of horizontal differencing, bit-plane reordering, and a final Zstandard compression stage, based off a framework from Mao et al. [5].

Due to the sheer size of the WSI cached for compression and the strain it will put on resources, the whole file is instead tiled into smaller images of size 1024×1024 . Each tile is compressed, decompressed, and then compared with the original tile by exact array equality before the result is accepted.

3.4 Metrics

To evaluate the performance of the algorithms (with and without TGTC), a range of metrics are considered to assess absolute size reduction, relative compression rate, and whether the method is lossless for tissue information. This study uses five metric families:

1. **Raw TGTC size reduction**

$$R_{\text{raw}} = \left(1 - \frac{S_{\text{TGTC}}}{S}\right) \times 100 \quad (1)$$

where S is the uncompressed `.dat` size and S_{TGTC} is the file size after TGTC pre-processing is applied to the corresponding baseline algorithm.

2. **Codec ratio on a given representation**

$$\rho_c(X) = \frac{X}{C_c(X)} \quad (2)$$

where $C_c(X)$ is the compressed size produced by codec c on data representation X of a given image.

3. **Final compressed-size reduction for a matched pair**

$$R_{\text{final},c} = \left(1 - \frac{C_c(X_{\text{TGTC}})}{C_c(X)}\right) \times 100 \quad (3)$$

where X_{TGTC} is the data representation of the pre-processed image with TGTC. This is the most important storage-facing metric because it directly compares the bytes that would be stored after compression.

4. **Tissue-retention metrics:** The fractions of retained tissue, discarded tissue, and removed background are captured to assess how much redundancy is in the image and how lossless the resultant compression actually is.
5. **Runtime metrics:** The time taken to compress, decompress, and the throughput (MB/s) are evaluated to help appreciate the impacts on real-world usage of any technique.

It is important to note the distinction between $p_c(X)$ and $R_{\text{final},c}$ is central to interpreting the results. A transformed image can be less compressible per byte than the original canvas while still yielding fewer final bytes overall because TGTC has already removed a substantial fraction of the canvas.

3.5 Tissue Retention Audit

In the absence of available pathologist-annotated ground truth masks for the slides used in this experiment, it was decided to utilise a classical thresholding method in the pipeline and audit it with a newer state of the art (SOTA) method. Ceachi et al (2025) introduced Double-Pass, an innovative unsupervised and annotation-free hybrid method for tissue detection for WSI’s [8]. This approach fuses two complementary strategies: *FilterGrays*, which utilises sharpening and colour-based artifact rejection, and *DownKMeans*, which performs clustering on downsampled RGB thumbnails to maximise speed. Double Pass will be used to detect tissue loss across all slides in the cohort to ensure robustness.

The audit file uses the same parameters as the authors original paper [8], while the threshold assigned to determine a PASS in the audit is set at 0.5% for the fraction of discarded tissue over all tissue on the whole slide outside of the crop footprint, the largest component fraction, or the largest discarded tissue fragment.

The full benchmark was executed on Kelvin2 (NI-HPC), using an NVidia A100 GPU. The benchmarking used compared all algorithms (with and without TGTC) against the four compression codecs and the tiled lossless verification path.

4 Results

4.1 TGTC Reduces Raw WSI Bytes Across the Full 12-Slide Cohort

Table 1 provides an overview of the size of the files before (baselines) and after the TGTC pre-processing of the image. In particular, the % reduction provides an overview of the amount of redundant background information that can be discarded through tissue identification, cropping, and rotation. Across the 12 paired slides, TGTC reduces aggregate raw storage from 145.40 GB to 114.36 GB, corresponding to an overall reduction of 21.35%. On a per-slide basis, the mean raw reduction is 29.81% and the median is 27.33%. The largest reduction is observed for brain (69.91%), while ovary shows the smallest but still positive raw reduction (6.48%). This means TGTC reduces raw storage burden on every slide in the current cohort before any codec is applied. Figure 1 illustrates each stage of the TGTC process including the final rotation, shown at Fig. 1(d), to orientate the image before saving in full resolution.

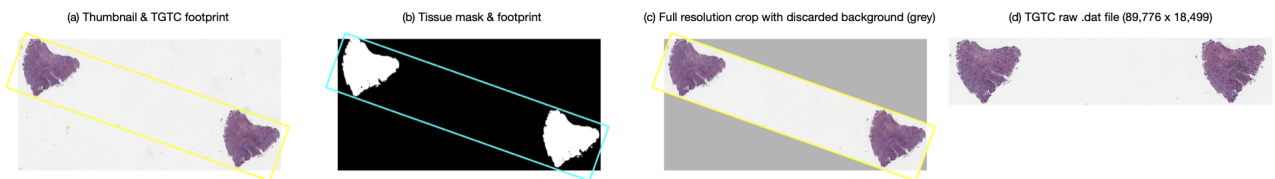


Figure 1: Representative TGTC process overlay for the lung slide. (a) The crop footprint is overlaid on the thumbnail and (b) tissue mask, (c) scaled to full resolution with discarded background shown in grey, and (d) the saved tissue is reserved after rotation.

Table 1: Paired baseline and TGTC .dat size and Raw Size Reduction (1) for the 12-slide HPC cohort.

Slide	Baseline (GB)	TGTC (GB)	Baseline Size	After TGTC	Size Reduction (%) (1)
Adrenal Gland	20.04	18.09	79,679 x 83,835	74,859 x 80,563	9.72
Bladder	21.00	15.39	10,1591 x 68,905	90,349 x 56,763	26.74
Brain	1.02	0.31	12,000 x 28,417	4,340 x 23,643	69.91
Breast	29.46	24.67	124,712 x 78,731	115,881 x 70,968	16.24
Cervix	2.34	1.92	24,940 x 31,295	20,877 x 30,717	17.84
Colon	10.61	6.04	55,040 x 64,256	43,432 x 46,372	43.05
Heart	4.61	3.08	32,671 x 47,076	21,036 x 48,856	33.18
Kidney	7.93	6.88	48,921 x 54,063	34,689 x 66,154	13.23
Lung	10.68	4.98	84,413 x 42,180	89,776 x 18,499	53.36
Ovary	28.79	26.93	109,559 x 87,606	107,792 x 83,270	6.48
Pancreas	5.97	4.30	49,880 x 39,867	33,348 x 42,986	27.91
Skin	2.94	1.76	27,888 x 35,169	21,925 x 26,790	40.11

4.2 Double-Pass Retention Audit find 99.94% Tissue-Preservation

While the double-pass retention audit shows there is some loss within the TGTC method, Table 2 shows all 12 audited slides PASS the audit and are safely inside the thresholds set, with mean retained tissue fraction 99.94%. The mean thumbnail-level removed area is 34.51% and the median is 35.31%, with only a very small fraction of this classified as discarded tissue and the vast majority being discarded background (See Figure 2). Despite positive results, the current audit cannot support a tissue-preserving losslessness claim for TGTC, however, it does provides strong evidence that TGTC achieves substantial data reduction while maintaining exceptionally high tissue retention, supporting its suitability for practical digital pathology workflows. Note: the ‘Area Removed (%)’ column from Table 2 and the ‘Size Reduction (%)’ column from Table 1 differ because while the former counts only discarded background pixels, excluding those pixels within the minimum bounding rectangle that lie outside of the original image, the latter measures the total pixel difference between the original and resized images.

Table 2: TGTC tissue-retention audit summary for the 12-slide HPC cohort.

Slide	Area Removed (%)	Retained Tissue (%)	Status
Adrenal Gland	10.83	99.62	PASS
Bladder	26.67	100.00	PASS
Brain	69.92	100.00	PASS
Breast	15.81	99.91	PASS
Cervix	26.62	100.00	PASS
Colon	46.69	100.00	PASS
Heart	38.19	100.00	PASS
Kidney	35.24	100.00	PASS
Lung	54.93	99.93	PASS
Ovary	11.19	100.00	PASS
Pancreas	35.37	99.81	PASS
Skin	42.61	100.00	PASS

4.3 Final Compressed Size Usually Improves Even With Less Redundant Data to Compress

When considering each compression method in combination and without the TGTC preprocessing block, we find that for each method, TGTC reduces the final compressed size in 40/48 cases (seen in Table 3), with mean improvement 10.39% and median improvement 6.55%. This is also strengthened with a small upward trend observed in the compression ratios, shown in Figure 3. Much of that work is to do with the removal of redundant background data during the preprocessing stage, as we often know that a transformed image contains fewer total bytes, but its remaining pixels are tissue-dense and therefore often less redundant than a large tissue-plus-background canvas for compression. Despite this, 40/48 files showed a positive improvement when TGTC is applied before compression. From a storage perspective, the final compressed size of a cohort matters more than individual slide representation where very small increases in size are negligible over full datasets. This is

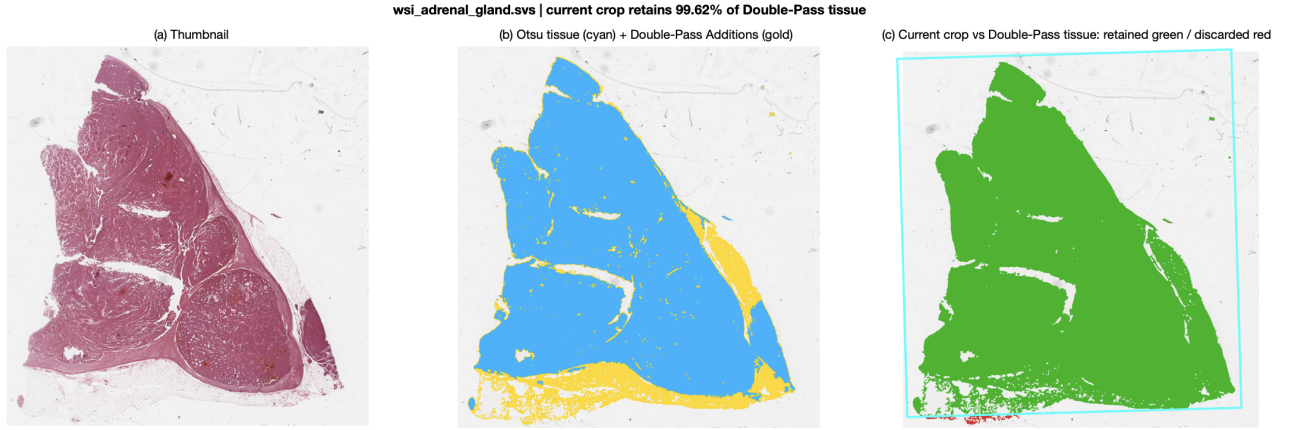


Figure 2: Double-Pass Audit for the Adrenal Gland slide - shows (a) original thumbnail, (b) Otsu’s tissue detection (in blue) with additional tissue detected by Double-Pass shown in gold and (c) tissue that was erroneously cropped by TGTC shown in red.

Table 3: Method-level summary over the 12-slide paired benchmark. Mean final reduction (%) is the average of the final compressed-size reduction for a matched pair ($R_{\text{final},c}$) for the same slide and codec. (3)

Method	Wins	Baseline Median Ratio ($\uparrow$)	TGTC Median Ratio ($\uparrow$)	Mean Final Reduction (%) ($\uparrow$)	Baseline Median Comp. Time (s $\downarrow$)	TGTC Median Comp. Time (s $\downarrow$)
JPEG-XL (tiled)	12/12	4.65	4.96	10.65	1629.41	1339.83
LZW (tiled)	12/12	1.79	1.90	11.39	82.35	53.51
WISE (Custom) (tiled)	10/12	3.16	3.31	10.82	63.33	40.55
ZStandard (tiled)	6/12	2.41	2.50	8.70	24.70	15.30

in addition to the storage savings already achieved on file size when preprocessing is applied. The results show that this storage-centric view is the right one for TGTC.

Method-level behaviour is also informative. LZW and JPEG-XL improve final compressed size on all 12 slides. WISE improves on 10 of 12 slides, and Zstandard improves on 6 of 12 slides. Zstandard nonetheless remains the fastest codec family in terms of compression throughput, illustrating a clear compression-ratio-versus-runtime trade-off.

4.4 Lost Tissue Data are Small and Localised

Table 4: Cases where TGTC increased final compressed size slightly.

Slide	Method	Final Reduction (%)
Ovary	ZStandard (tiled)	-2.78
Skin	ZStandard (tiled)	-2.05
Adrenal Gland	ZStandard (tiled)	-0.87
Ovary	WISE (Custom) (tiled)	-0.79
Colon	ZStandard (tiled)	-0.58
Skin	WISE (Custom) (tiled)	-0.45
Bladder	ZStandard (tiled)	-0.44
Breast	ZStandard (tiled)	-0.17

There are eight negative paired comparisons in the full benchmark (Table 4), all relatively small in magnitude. Six occur with Zstandard and two with WISE. The nature of the Zstandard and WISE algorithms may mean that there is an optimal orientation of the image to best utilise the style of compression which does not necessarily mean reducing the volume of information to be compressed. The worst compression is seen in the ovary sample using Zstandard where final compressed size increases by 2.78%. These negative results are important to highlight as they show that TGTC is not uniformly beneficial for every slide-codec combination.

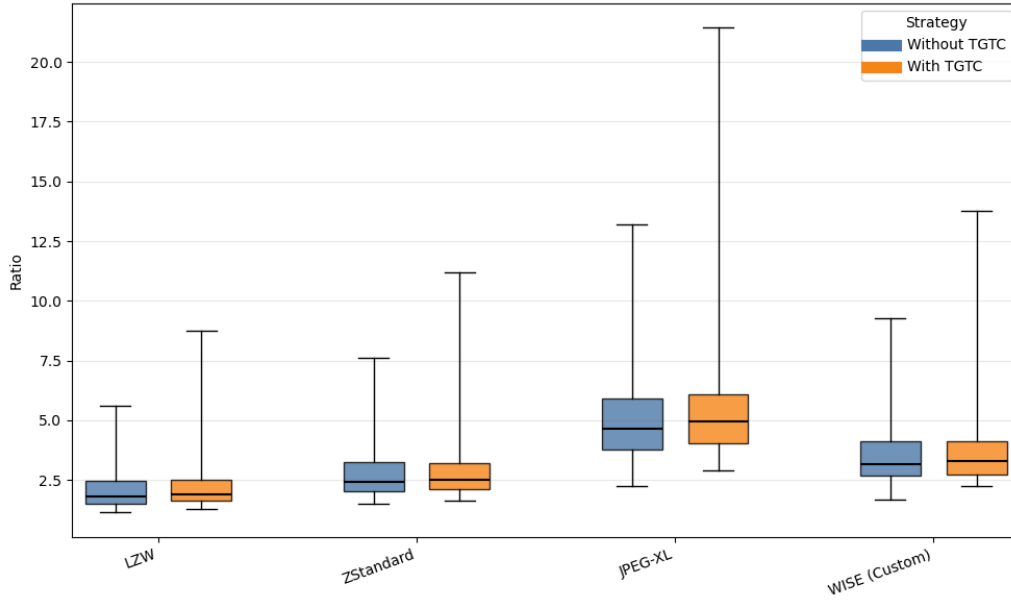


Figure 3: Box-and-whisker comparison of codec ratios for baseline and TGTC-transformed inputs across the 12-slide cohort in tiled format. TGTC generally shifts ratio trend upwards to improve final compressed size overall, with marginal increases in the median ratio shown in every compression method and similar patterns within the interquartile range with slight upward trends.

At the same time, the overall pattern remains favourable: all raw pairs shrink, all slides pass the retention audit, and most final compressed results improve.

4.5 Runtime Explains the Practical Limit on Dataset Size

The HPC benchmark clarifies why WSI compression studies are costly to scale. Median baseline compression times are 24.7s for Zstandard, 82.4s for LZW, 63.3s for WISE, and 1629.4s for JPEG-XL. On TGTC inputs, the corresponding medians fall respectively to 15.3s, 53.5s, 40.6s, and 1339.8s for each of the compression methods. Median baseline compression throughput is 464.5MB/s for Zstandard, 143.3MB/s for WISE, 108.4MB/s for LZW, and 4.3MB/s for JPEG-XL.

These numbers make the practical trade-off visible. JPEG-XL gives the highest compression ratios, but it is far slower than the other methods. Zstandard is consistently the fastest, but its storage benefit after TGTC is less reliable than LZW or JPEG-XL. WISE occupies a middle region: it is much faster than JPEG-XL while typically retaining a larger storage gain than Zstandard.

5 Discussion

This study provides encouraging evidence that TGTC can reduce whole-slide image storage requirements while maintaining a high level of tissue preservation. Four key findings emerge from the evaluation. First, **TGTC consistently reduced raw background burden** across the dataset, with every slide demonstrating a positive reduction in uncropped file size and an overall storage reduction of 21.35%. Second, **tissue-retention analysis showed a high degree of preservation**, with all 12 slides passing the SOTA Double-Pass tissue audit and achieving a mean retained tissue fraction of 99.94%. Third, **the effectiveness of TGTC is influenced by the choice of compression codec**. Storage improvements were observed in 40/48 paired comparisons across four codecs, with LZW and JPEG-XL producing file-size reductions for every slide in the cohort, although the magnitude of benefit remained codec dependent. Finally, **computational cost remains a practical consideration**. Even with serial HPC execution, benchmarking a 12-slide cohort across four codecs required substantial compute resources, particularly for JPEG-XL, which may limit larger-scale evaluations.

Despite these promising findings, several limitations should be acknowledged. Tissue-preservation was assessed using thumbnail-level audit masks rather than full-resolution pathological review, and future studies should incorporate pathologist validation to establish clinical relevance. While TGTC generally improved storage efficiency, results remain dependent on codec selection, compression settings, and tile configuration. The Double-Pass Tissue Audit also revealed minor loss of faint tissue structures along some tissue boundaries following cropping and rotation. Although retention remained well within acceptable thresholds, clinical significance of these losses are unknown and warrants further investigation. Refinement of the Otsu thresholding stage may help improve tissue identification and reduce the risk of inadvertent tissue exclusion. Finally, conclusions are based on a relatively small cohort of 12 WSI's. Future work should evaluate TGTC on larger, more diverse datasets to assess the generalisability of observed storage and tissue-retention performance.

6 Conclusion

This study demonstrates that Thumbnail-Guided Tissue Compaction (TGTC) is a practical preprocessing stage for WSI storage pipelines. As shown in Section 4.1, TGTC reduced aggregate raw storage requirements from 145.40GB to 114.36GB across the 12-slide cohort. In digital pathology environments, such reductions could lower storage costs, reduce transmission burdens, and improve the scalability of growing slide archives.

The compression benchmarking results in Section 4.3 show that TGTC improved the final compressed file size in 40 of 48 paired slide-codec comparisons, indicating that TGTC can complement existing compression approaches without requiring changes to downstream workflows. Tissue-retention analysis (Section 4.2) further showed that all audited slides passed the Double-Pass Tissue Audit, with a mean retained tissue fraction of 99.94%, suggesting that meaningful storage savings can be achieved while preserving tissue content under the current audit methodology. Among the evaluated codecs, LZW and JPEG-XL produced the most consistent benefits when combined with TGTC. JPEG-XL achieved the strongest compression performance, making it attractive for archival storage, while LZW offered a more favourable balance between storage reduction and processing time for active use.

Overall, the results suggest that TGTC offers a promising route towards reducing the growing storage burden of digital pathology while maintaining high levels of tissue retention. Future work should validate tissue preservation using higher-resolution and task-based assessments and evaluate performance across larger and more diverse datasets.

References

- [1] X. Matias-Guiu, J. Temprana-Salvador, P. G. Lopez, S. F. Kammerer-Jacquet, N. Rioux-Leclercq, D. Clark, C. M. Schürch, F. Fend, S. Mattern, D. Snead, N. Fusco, E. Guerini-Rocco, F. Rojo, M. Brevet, M. S. Tellez, A. D. Tos, T. di Maio, S. Ramírez-Peinado, E. Sheppard, H. Bannister, A. Gkiokas, M. Arpaia, O. B. Dhia, and N. Martino, "Implementing digital pathology: qualitative and financial insights from eight leading european laboratories," *Virchows Archiv*, 2025.
- [2] T. A. Welch, "A technique for high-performance data compression," *Computer*, vol. 17, no. 6, pp. 8–19, 1984.
- [3] Y. Collet and M. Kucherawy, "RFC 8878: Zstandard Compression and the 'application/zstd' Media Type," IETF, Tech. Rep., 2021. [Online]. Available: <https://www.rfc-editor.org/info/rfc8878>
- [4] J. Alakuijala, R. van Asseldonk, S. Boukott, M. Bruse, I.-M. Comsa, M. Firsching, T. Fischbacher, S. Gomez, E. Kliuchnikov, R. Obryk, K. Potempa, A. Rhatushnyak, J. Sneyers, Z. Szabadka, L. Vandevenne, L. Versari, and J. Wassenberg, "JPEG XL next-generation image compression architecture and coding tools," in *Applications of Digital Image Processing XLIII*. SPIE, 2019. [Online]. Available: <https://research.google/pubs/jpeg-xl-next-generation-image-compression-architecture-and-coding-tools/>
- [5] Y. Mao, J. Wang, N. Guan, and C. J. Xue, "Wise: A framework for gigapixel whole-slide-image lossless compression," 2025. [Online]. Available: <https://arxiv.org/abs/2503.18074>
- [6] N. C. I. USA, "<https://www.cancer.gov/tcga>."
- [7] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.
- [8] B. Ceachi, F. Muresan, M. Trascau, and A. M. Florea, "Efficient tissue detection in whole-slide images using classical and hybrid methods: Benchmark on tcga cancer cohorts," *Cancers*, vol. 17, no. 17, 2025. [Online]. Available: <https://www.mdpi.com/2072-6694/17/17/2918>